# CP-violating Type-II 2HDM under Soft Broken $\mathbb{Z}_2$ Symmetry and Parameters around SM Alignment Point

Shangchengze Li
*Advisor: Prof. Yusheng Wu*
School of the Gifted Young, University of Science and Technology of China

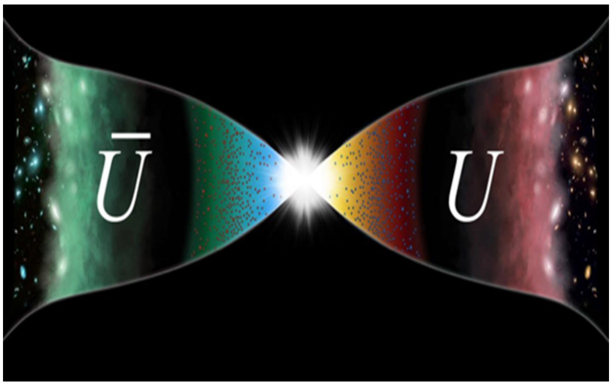
December 2025

**Abstract**

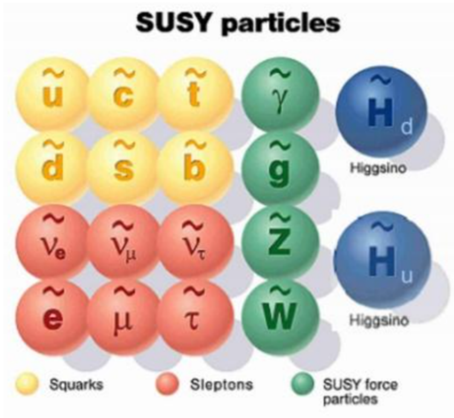
This report investigates the CP-violating Type-II Two-Higgs-Doublet Model (C2HDM) as a solution to the insufficient CP violation in the Standard Model. We utilize `C2HDM_HDECAY` and `HiggsTools` to analyze the parameter space around the SM alignment point. By performing detailed scans on mixing angles $(\alpha_1, \alpha_2, \alpha_3)$ and masses $(m_{H_2}, m_{H^\pm})$, we evaluate the compatibility with current LHC 125 GeV Higgs experiment data. The study reveals that the SM alignment limit is not strictly the global best-fit point especially for $\alpha_1$ and $\alpha_2 (tan\beta < 1)$, with sensitivities strictly dependent on $\tan\beta$.

## 1 Background

The Standard Model (SM) does not provide enough CP violation to achieve the observed matter-antimatter asymmetry, thus necessitating the exploration of CP violation sources beyond the SM. The Two-Higgs-Doublet Model (2HDM) is one of the simplest extensions capable of providing new CP violation sources.




(a) Matter-antimatter Asymmetry.




(b) Connection to SUSY.

Figure 1: Background motivation: Asymmetry problem and SUSY connection.

Specifically, the Type-II Yukawa coupling in 2HDM forces u-type quarks and d-type quarks to couple with different Higgs complex doublets. This structure is phenomenologically identical to the Yukawa coupling in the Minimal Supersymmetric Standard Model (MSSM) with respect to Higgs doublets $H_u$ and $H_d$.

## 2 General 2HDM and $\mathbb{Z}_2$ Symmetry

Here we present the general Lagrangian of 2HDM, consisting of the scalar potential and Yukawa interaction terms:

$$\begin{aligned}\mathcal{L}_{2HDM} &= m_{11}^2\Phi_1^\dagger\Phi_1 + m_{22}^2\Phi_2^\dagger\Phi_2 + (m_{12}^2\Phi_1^\dagger\Phi_2 + h.c.)\\ &+ \lambda_1(\Phi_1^\dagger\Phi_1)^2 + \lambda_2(\Phi_2^\dagger\Phi_2)^2 + \lambda_3\Phi_1^\dagger\Phi_1\Phi_2^\dagger\Phi_2 + \lambda_4\Phi_1^\dagger\Phi_2\Phi_2^\dagger\Phi_1\\ &+ \Big(\lambda_5(\Phi_1^\dagger\Phi_2)^2 + h.c.\Big) + \Phi_1^\dagger\Phi_1(\lambda_6\Phi_1^\dagger\Phi_2 + h.c.) + \Phi_2^\dagger\Phi_2(\lambda_7\Phi_1^\dagger\Phi_2 + h.c.)\end{aligned} \tag{1}$$

$$\mathcal{L}_{Yukawa} = \bar{Q}_L Y_1^u\tilde{\Phi}_1 u_R + \bar{Q}_L Y_2^u\tilde{\Phi}_2 u_R + \bar{Q}_L Y_1^d\Phi_1 d_R + \bar{Q}_L Y_2^d\Phi_2 d_R + \bar{L}_L Y_1^e\Phi_1 e_R + \bar{L}_L Y_2^e\Phi_2 e_R + h.c. \tag{2}$$

(Note: For the Yukawa terms, $\tilde{\Phi}_{1,2} = i\sigma_2\Phi_{1,2}^*$.)

In a general 2HDM with two complex Higgs doublets $\Phi_1$ and $\Phi_2$, we typically cannot diagonalize Yukawa coupling matrices $Y_1^x$ and $Y_2^x$ simultaneously, which leads to Flavor Changing Neutral Currents (FCNCs). To avoid this, a $\mathbb{Z}_2$ symmetry and a corresponding $\mathbb{Z}_2$ charge are introduced.

| | $\Phi_1$ | $\Phi_2$ | $u_R$ | $d_R$ | $e_R$ | $Q_L$, $L_L$ |
|---|---|---|---|---|---|---|
| Type-I | + | − | − | − | − | + |
| Type-II | + | − | − | + | + | + |
| Type-X | + | − | − | − | + | + |
| Type-Y | + | − | − | + | − | + |

(a) Z2 charges assignment to avoid FCNCs.

| | $u$-type | $d$-type | leptons |
|---|---|---|---|
| Type I | $\Phi_2$ | $\Phi_2$ | $\Phi_2$ |
| Type II | $\Phi_2$ | $\Phi_1$ | $\Phi_1$ |
| Lepton-Specific | $\Phi_2$ | $\Phi_2$ | $\Phi_1$ |
| Flipped | $\Phi_2$ | $\Phi_1$ | $\Phi_2$ |

(b) Yukawa couplings structure.

Figure 2: Z2 charges and Yukawa structure in different type of 2HDM.

The general potential is given by:

$$\begin{aligned}\mathcal{L}_{2HDM} &= m_{11}^2\Phi_1^\dagger\Phi_1 + m_{22}^2\Phi_2^\dagger\Phi_2 - (m_{12}^2\Phi_1^\dagger\Phi_2 + h.c.)\\ &+ \lambda_1(\Phi_1^\dagger\Phi_1)^2 + \lambda_2(\Phi_2^\dagger\Phi_2)^2 + \lambda_3\Phi_1^\dagger\Phi_1\Phi_2^\dagger\Phi_2\\ &+ \lambda_4\Phi_1^\dagger\Phi_2\Phi_2^\dagger\Phi_1 + (\frac{\lambda_5}{2}(\Phi_1^\dagger\Phi_2)^2 + h.c.) + \ldots\end{aligned} \tag{3}$$

## 3 Type-II C2HDM Framework

We focus on the **C2HDM**, defined as CP-violating 2HDM under Soft Broken $\mathbb{Z}_2$ Symmetry. The scalar potential is given by:

$$\begin{aligned}\mathcal{L}_{2HDM} &= m_{11}^2\Phi_1^\dagger\Phi_1 + m_{22}^2\Phi_2^\dagger\Phi_2 - (m_{12}^2\Phi_1^\dagger\Phi_2 + h.c.)\\ &+ \lambda_1(\Phi_1^\dagger\Phi_1)^2 + \lambda_2(\Phi_2^\dagger\Phi_2)^2 + \lambda_3\Phi_1^\dagger\Phi_1\Phi_2^\dagger\Phi_2\\ &+ \lambda_4\Phi_1^\dagger\Phi_2\Phi_2^\dagger\Phi_1 + (\frac{\lambda_5}{2}(\Phi_1^\dagger\Phi_2)^2 + h.c.)\end{aligned} \tag{4}$$

In the potential, parameters $m_{12}^2$ and $\lambda_5$ are complex, while the rest are real. The scalar doublets are expanded as:

$$\Phi_1 = \begin{pmatrix}\phi_1^+\\ \frac{v_1+\rho_1+i\eta_1}{\sqrt{2}}\end{pmatrix}, \quad \Phi_2 = \begin{pmatrix}\phi_2^+\\ \frac{v_2+\rho_2+i\eta_2}{\sqrt{2}}\end{pmatrix} \tag{5}$$

We define $\tan\beta = v_2/v_1$ and $v = \sqrt{v_1^2 + v_2^2} \approx 246$ GeV.

### 3.1 Higgs Basis and Scalar Mixing Angles

In the Higgs Basis, the fields are rotated:

$$\begin{pmatrix} \mathcal{H}_1 \\ \mathcal{H}_2 \end{pmatrix} = \begin{pmatrix} c_\beta & s_\beta \\ -s_\beta & c_\beta \end{pmatrix} \begin{pmatrix} \Phi_1 \\ \Phi_2 \end{pmatrix} \tag{6}$$

where $\mathcal{H}_1$ contains the SM VEV $v$. The physical mass eigenstates $H_1, H_2, H_3$ are obtained by diagonalizing the neutral components using scalar mixing angles $0 < \alpha_1, \alpha_2, \alpha_3 < \pi/2$:

$$\begin{pmatrix} H_1 \\ H_2 \\ H_3 \end{pmatrix} = R(\alpha_1, \alpha_2, \alpha_3) \begin{pmatrix} \rho_1 \\ \rho_2 \\ \rho_3 \end{pmatrix} \tag{7}$$

Here, $\rho_3 = -s_\beta \eta_1 + c_\beta \eta_2$. The rotation matrix $R$ is explicitly given by:

$$R = \begin{pmatrix} c_1 c_2 & s_1 c_2 & s_2 \\ -(c_1 s_2 s_3 + s_1 c_3) & c_1 c_3 - s_1 s_2 s_3 & c_2 s_3 \\ -c_1 s_2 c_3 + s_1 s_3 & -(c_1 s_3 + s_1 s_2 c_3) & c_2 c_3 \end{pmatrix} \tag{8}$$

(where $c_i \equiv \cos\alpha_i$ and $s_i \equiv \sin\alpha_i$).

And we have the mass relation:

$$m_{H_3}^2 = \frac{m_{H_1}^2 R_{13}(R_{12} t_\beta - R_{11}) + m_{H_2}^2 R_{23}(R_{22} t_\beta - R_{21})}{R_{33}(R_{31} - R_{32} t_\beta)} \tag{9}$$

There are 9 independent parameters in C2HDM:

$$v, \quad m_{H_1}, \quad m_{H_2}, \quad m_{H^\pm}, \quad \alpha_1, \quad \alpha_2, \quad \alpha_3, \quad \tan\beta, \quad \mathrm{Re}[m_{12}^2] \tag{10}$$

We set $m_{H_1} = 125$ GeV (SM-like) and assume $m_{H_1} < m_{H_2} < m_{H_3} < 1000$ GeV.

## 4 Effective Couplings Modification

The modification factors for the Yukawa couplings of the scalar mass eigenstates $H_i$ $(i = 1, 2, 3)$ to fermions relative to the SM Higgs couplings are given in the table below. Note that $s_\beta \equiv \sin\beta$, $c_\beta \equiv \cos\beta$, and $t_\beta \equiv \tan\beta$.

| | $u$-type | $d$-type | leptons |
|---|---|---|---|
| Type I | $\frac{R_{i2}}{s_\beta} - i\frac{R_{i3}}{t_\beta}\gamma_5$ | $\frac{R_{i2}}{s_\beta} + i\frac{R_{i3}}{t_\beta}\gamma_5$ | $\frac{R_{i2}}{s_\beta} + i\frac{R_{i3}}{t_\beta}\gamma_5$ |
| Type II | $\frac{R_{i2}}{s_\beta} - i\frac{R_{i3}}{t_\beta}\gamma_5$ | $\frac{R_{i1}}{c_\beta} - i t_\beta R_{i3}\gamma_5$ | $\frac{R_{i1}}{c_\beta} - i t_\beta R_{i3}\gamma_5$ |
| Lepton-Specific | $\frac{R_{i2}}{s_\beta} - i\frac{R_{i3}}{t_\beta}\gamma_5$ | $\frac{R_{i2}}{s_\beta} + i\frac{R_{i3}}{t_\beta}\gamma_5$ | $\frac{R_{i1}}{c_\beta} - i t_\beta R_{i3}\gamma_5$ |
| Flipped | $\frac{R_{i2}}{s_\beta} - i\frac{R_{i3}}{t_\beta}\gamma_5$ | $\frac{R_{i1}}{c_\beta} - i t_\beta R_{i3}\gamma_5$ | $\frac{R_{i2}}{s_\beta} + i\frac{R_{i3}}{t_\beta}\gamma_5$ |

The coupling of the scalar $H_i$ to vector bosons ($VV$, where $V = W, Z$), relative to the SM coupling, is given by:

$$c(H_i VV) = c_\beta R_{i1} + s_\beta R_{i2} \tag{11}$$

For the lightest neutral scalar $H_1$ in the Type-II model, the squared moduli of the Yukawa couplings to bottom and top quarks are:

$$|c(H_1 b\bar{b})|^2 = \frac{c_1^2 c_2^2}{c_\beta^2} + t_\beta^2 s_2^2 \tag{12}$$

$$|c(H_1 t\bar{t})|^2 = \frac{s_1^2 c_2^2}{s_\beta^2} + \frac{s_2^2}{t_\beta^2} \tag{13}$$

The gauge coupling for $H_1$ can be expressed in terms of the mixing angles as:

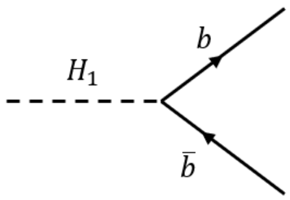
$$c(H_1 VV) = c_2 \cos(\beta - \alpha_1) \tag{14}$$

The partial decay width for the SM-like Higgs boson $h$ (identified as $H_1$) into two photons includes contributions from fermion loops, the $W$ boson loop, and the charged Higgs $H^\pm$ loop [1]:

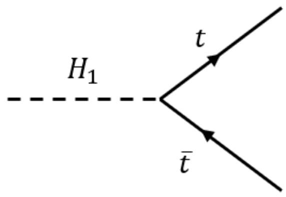
$$\Gamma(h \to \gamma\gamma) = \frac{G_F \alpha^2 M_h^3}{128\sqrt{2}\pi^3} \left| \sum_{f \in \{t,b\}} g_f Q_f^2 N_c A_{1/2}(\tau_f) + g_W A_1(\tau_W) + g_h A_0(\tau_{H^\pm}) \right|^2 \tag{15}$$

where $\tau_i$, $g_h$ and $A_0(\tau)$ are defined as:

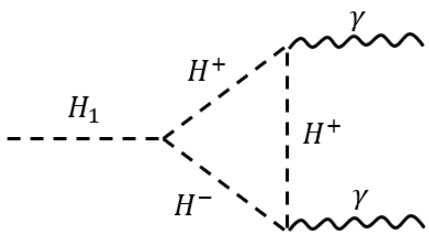
$$\tau_i = \frac{M_h^2}{4m_i^2}, \quad g_h = -\frac{m_W}{g m_{H^\pm}^2} g_{hH^+H^-}, \quad A_0(\tau) = -[\tau - f(\tau)]\tau^{-2} \tag{16}$$




(a) $H_1 bb$ vertex




(b) $H_1 tt$ vertex




(c) the charged-Higgs loop Feynman diagram for $H_1 \to \gamma\gamma$

Figure 3: Effective coupling vertices and loop diagram.

# 5 Numerical Analysis Tools

## 5.1 C2HDM_HDECAY

We use `C2HDM_HDECAY` [2], which is the implementation of C2HDM in `HDECAY v6.51`.

- Calculates all decay channels kinematically allowed with branching ratios $> 10^{-4}$.
- Includes state-of-the-art higher-order QCD corrections and relevant off-shell decays (excluding Higgs-to-Higgs).
- Electroweak corrections are not included.

### 5.2 HiggsTools

We employ `HiggsTools` [3], which includes:

- **HiggsPredictions (HP): defines the BSM particles and obtains theory prediction for production and decay rates. Also it can calculate the production cross sections of BSM particle briefly by effective couplings modification input.
- **HiggsBounds (HB): evaluates bounds from direct searches for BSM particles by CLs method; it will calculate expected up limits(95% C.L.) and observed up limits(95% C.L.) for every relevant process with respect to every BSM particle; (the HiggsBounds limit database contains 258 different experimental limits from LEP and the LHC.)
- **HiggsSignals (HS): evaluates compatibility with the measurements of the Higgs boson detected at 125 GeV by calculating $\chi^2$ with experimental results with respect to the 125GeV signal. Now we can only input inclusive cross sections and branch ratios, but HiggsSignals compute $\chi^2$ with different bins divided by different experimental reports using STXS method. So, it uses the SM Higgs boson as a kinematic template, and provide you a choose to input the modification factors for diferent bins. But unfortunately, we can't get the factors without Monte Carlo simulation.

All the measurements used by HiggsSignals to compute $\chi^2$ comes from 21 experiment reports about:

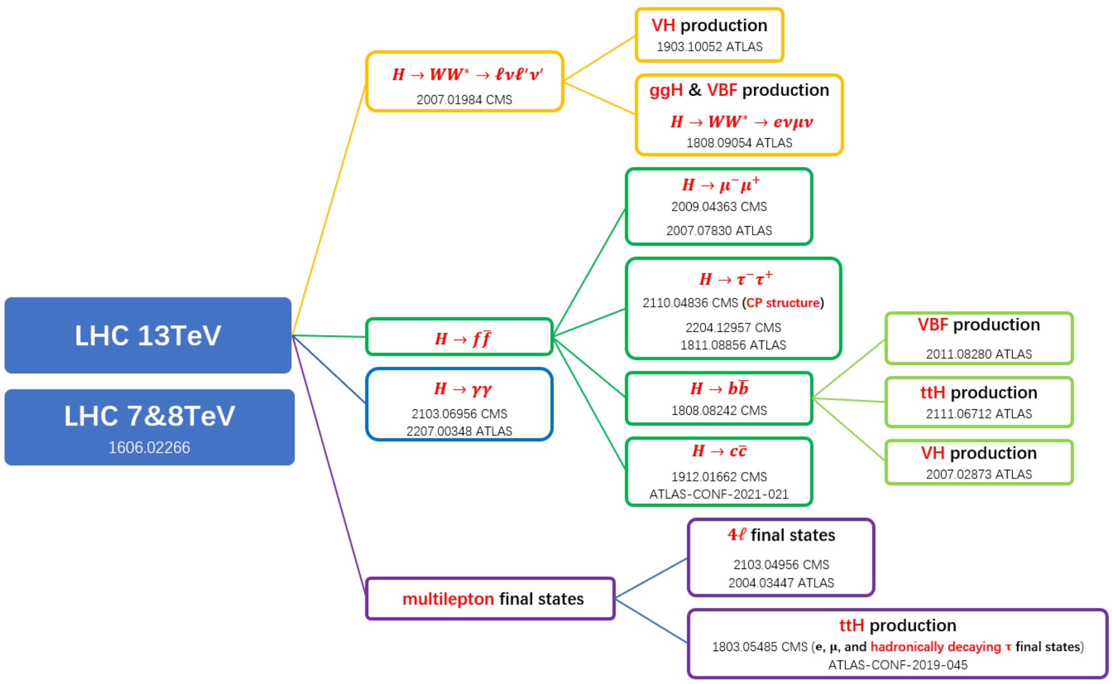



Figure 4: All the measurements used by HiggsSignals

# 6 SM Alignment and Parameter Impacts on $\chi^2$

### 6.1 SM Alignment at tree-level

SM alignment means we should create a scalar particle that is almost the same as SM Higgs boson, and other scalar particles have negligible impacts on it.

We set up $H_1$ as the 125 GeV SM-like CP-even Higgs. The alignment limit requires all effective coupling modification factors to be 1. This implies:

$$\alpha_1 = \beta, \quad \alpha_2 = 0 \tag{17}$$

Under this condition,

$$|c(H_1VV)|^2 = |c(H_1b\bar{b})|^2 = |c(H_1t\bar{t})|^2 = 1 \tag{18}$$

$$|c(H_2VV)|^2 = |c(H_3VV)|^2 = 0 \tag{19}$$

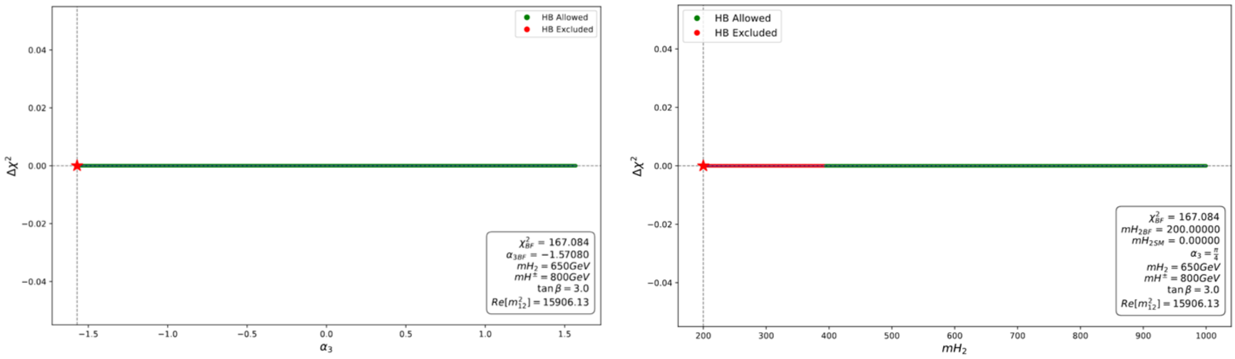
$$|c(H_2H_1Z)|^2 = |c(H_3H_1Z)|^2 = |c(H^{\pm}W^{\mp}H_1)|^2 = 0 \tag{20}$$

$$m_{H_2} = m_{H_3}(\alpha_2 = 0) \tag{21}$$

### 6.2 Impact of $\alpha_3$ and $m_{H_2}$ on $\chi^2$

Using HiggsSignals, we scan $\alpha_3$ and $m_{H_2}$ to see which parameter region has smaller impacts on $\chi^2$.

First, for $\alpha_3$ and $m_{H_2}$, We compute the $\chi^2$ and show the figure of $\Delta\chi^2$, which means $\chi^2 - \chi^2_{min}$.




(a) Impact of $\alpha_3$. (b) Impact of $m_{H_2}$.

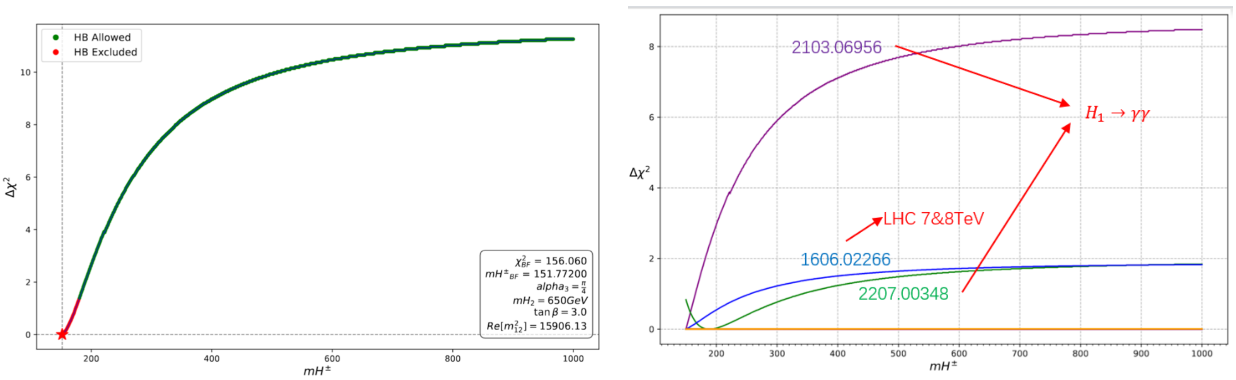
Figure 5: Scan results for $\alpha_3$ and $m_{H_2}$.

According to Figure 5(a), $\Delta\chi^2$ remains constant as $\alpha_3$ varies, and none of the points are excluded by HiggsBounds, so we can choose $\alpha_3$ freely. According to Figure 5(b), $m_{H_2}$ almost has the same situation, except for light-mass region which is excluded by HiggsBounds,so we can choose a heavy $m_{H_2}$.

So finally, we set $\alpha_3 = \pi/4$ and $m_{H_2} = 650$ GeV for further study.

### 6.3 Impact of $m_{H^{\pm}}$ on $\chi^2$ and Decoupling limit

We perform a brief scan on $m_{H^{\pm}}$.

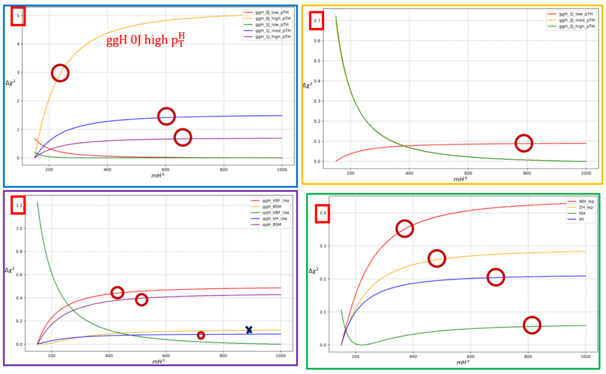



(a) Global $\Delta\chi^2$. (b) Partial $\Delta\chi^2$ with different measurements.

Figure 6: Impact of charged Higgs mass on Global and Partial$\chi^2$.

As seen in Figure 6(a), $m_{H^\pm}$ has a significant impact on $\chi^2$. And from Figure 6(b), we knew it's mainly because $m_{H^\pm}$ influences $\Gamma(H_1 \to \gamma\gamma)$. But actually, HiggsSignals computes $\chi^2$ with experimental results instead of SM predictions. So the BFP in the figure doesn't mean SM Alignment.

So let's check the $\Delta\chi^2$ computed with arXiv:2103.06956 [4].




(a) Partial $\Delta\chi^2$ for every bins.

| Parameters | SM prediction ($m_H = 125.38$ GeV) | $\sigma\mathcal{B}$ (fb) Observed (Expected) Best fit | Stat. unc. | Syst. unc. | $\sigma\mathcal{B}/(\sigma\mathcal{B})_{SM}$ Observed (Expected) Best fit |
|---|---|---|---|---|---|
| ggH 0J low $p_T^H$ | $15.21^{+4.14}_{-4.18}$ | $9.41^{+3.92}_{-3.99}\left(^{+4.20}_{-4.06}\right)$ | $^{+3.90}_{-3.98}\left(^{+4.16}_{-4.05}\right)$ | $^{+0.44}_{-0.25}\left(^{+0.51}_{-0.33}\right)$ | $0.62^{+0.26}_{-0.26}\left(^{+0.28}_{-0.27}\right)$ |
| ggH 0J high $p_T^H$ | $44.25^{+4.84}_{-4.61}$ | $58.50^{+8.10}_{-7.17}\left(^{+7.87}_{-7.77}\right)$ | $^{+7.70}_{-6.91}\left(^{+7.67}_{-7.63}\right)$ | $^{+2.50}_{-1.92}\left(^{+1.78}_{-1.42}\right)$ | $1.32^{+0.18}_{-0.16}\left(^{+0.18}_{-0.18}\right)$ |
| ggH 1J low $p_T^H$ | $16.20^{+2.25}_{-2.27}$ | $13.39^{+5.58}_{-5.49}\left(^{+5.67}_{-5.59}\right)$ | $^{+5.52}_{-5.45}\left(^{+5.61}_{-5.56}\right)$ | $^{+0.80}_{-0.63}\left(^{+0.77}_{-0.48}\right)$ | $0.83^{+0.34}_{-0.34}\left(^{+0.35}_{-0.34}\right)$ |
| ggH 1J med $p_T^H$ | $11.23^{+1.56}_{-1.55}$ | $13.66^{+2.91}_{-2.96}\left(^{+3.15}_{-3.39}\right)$ | $^{+2.83}_{-2.92}\left(^{+3.09}_{-3.36}\right)$ | $^{+0.70}_{-0.50}\left(^{+0.59}_{-0.45}\right)$ | $1.22^{+0.26}_{-0.26}\left(^{+0.28}_{-0.30}\right)$ |
| ggH 1J high $p_T^H$ | $2.00^{+0.36}_{-0.36}$ | $2.56^{+0.90}_{-0.87}\left(^{+0.91}_{-0.92}\right)$ | $^{+0.90}_{-0.87}\left(^{+0.90}_{-0.90}\right)$ | $^{+0.11}_{-0.11}\left(^{+0.15}_{-0.19}\right)$ | $1.28^{+0.45}_{-0.44}\left(^{+0.46}_{-0.46}\right)$ |
| ggH ≥2J low $p_T^H$ | $2.82^{+0.68}_{-0.68}$ | $3.62^{+3.65}_{-3.55}\left(^{+3.73}_{-2.82}\right)$ | $^{+3.62}_{-3.53}\left(^{+3.69}_{-2.82}\right)$ | $^{+0.41}_{-0.31}\left(^{+0.55}_{-0.55}\right)$ | $1.29^{+1.29}_{-1.26}\left(^{+1.32}_{-1.00}\right)$ |
| ggH ≥2J med $p_T^H$ | $4.53^{+1.07}_{-1.07}$ | $0.08^{+2.77}_{-0.08}\left(^{+2.87}_{-2.82}\right)$ | $^{+2.76}_{-0.08}\left(^{+2.84}_{-2.82}\right)$ | $^{+0.28}_{-0.08}\left(^{+0.38}_{-0.14}\right)$ | $0.02^{+0.61}_{-0.02}\left(^{+0.63}_{-0.62}\right)$ |
| ggH ≥2J high $p_T^H$ | $2.12^{+0.49}_{-0.50}$ | $0.82^{+0.92}_{-0.82}\left(^{+1.15}_{-1.10}\right)$ | $^{+0.88}_{-0.82}\left(^{+1.11}_{-1.09}\right)$ | $^{+0.26}_{-0.26}\left(^{+0.31}_{-0.14}\right)$ | $0.39^{+0.43}_{-0.39}\left(^{+0.54}_{-0.52}\right)$ |
| ggH VBF-like | $2.22^{+0.52}_{-0.52}$ | $5.86^{+2.45}_{-2.59}\left(^{+2.90}_{-2.22}\right)$ | $^{+2.27}_{-2.55}\left(^{+2.81}_{-2.22}\right)$ | $^{+0.92}_{-0.48}\left(^{+0.71}_{-0.71}\right)$ | $2.64^{+1.10}_{-1.17}\left(^{+1.31}_{-1.00}\right)$ |
| ggH BSM | $1.43^{+0.36}_{-0.35}$ | $1.34^{+0.50}_{-0.47}\left(^{+0.59}_{-0.49}\right)$ | $^{+0.49}_{-0.46}\left(^{+0.58}_{-0.49}\right)$ | $^{+0.05}_{-0.09}\left(^{+0.09}_{-0.05}\right)$ | $0.94^{+0.35}_{-0.33}\left(^{+0.41}_{-0.35}\right)$ |
| qqH VBF-like | $2.96^{+0.59}_{-0.59}$ | $0.49^{+1.44}_{-0.49}\left(^{+1.49}_{-1.53}\right)$ | $^{+1.40}_{-0.49}\left(^{+1.47}_{-1.47}\right)$ | $^{+0.34}_{-0.34}\left(^{+0.25}_{-0.43}\right)$ | $0.17^{+0.49}_{-0.17}\left(^{+0.50}_{-0.52}\right)$ |
| qqH VH-like | $1.22^{+0.05}_{-0.04}$ | $1.57^{+1.20}_{-1.24}\left(^{+1.15}_{-1.23}\right)$ | $^{+1.19}_{-1.21}\left(^{+1.15}_{-1.23}\right)$ | $^{+0.13}_{-0.26}\left(^{+0.07}_{-0.04}\right)$ | $1.29^{+0.98}_{-1.01}\left(^{+0.94}_{-1.01}\right)$ |
| qqH BSM | $0.37^{+0.03}_{-0.02}$ | $0.52^{+0.24}_{-0.22}\left(^{+0.26}_{-0.23}\right)$ | $^{+0.24}_{-0.22}\left(^{+0.25}_{-0.23}\right)$ | $^{+0.03}_{-0.01}\left(^{+0.03}_{-0.01}\right)$ | $1.42^{+0.65}_{-0.59}\left(^{+0.69}_{-0.62}\right)$ |
| WH lep | $0.88^{+0.03}_{-0.03}$ | $1.19^{+0.49}_{-0.44}\left(^{+0.51}_{-0.42}\right)$ | $^{+0.48}_{-0.43}\left(^{+0.50}_{-0.41}\right)$ | $^{+0.07}_{-0.04}\left(^{+0.05}_{-0.05}\right)$ | $1.35^{+0.55}_{-0.49}\left(^{+0.57}_{-0.47}\right)$ |
| ZH lep | $0.54^{+0.03}_{-0.02}$ | $0.71^{+0.41}_{-0.35}\left(^{+0.42}_{-0.35}\right)$ | $^{+0.40}_{-0.35}\left(^{+0.41}_{-0.35}\right)$ | $^{+0.07}_{-0.03}\left(^{+0.06}_{-0.03}\right)$ | $1.32^{+0.76}_{-0.65}\left(^{+0.78}_{-0.65}\right)$ |
| $t\bar{t}$H | $1.13^{+0.08}_{-0.11}$ | $1.13^{+0.42}_{-0.39}\left(^{+0.42}_{-0.41}\right)$ | $^{+0.42}_{-0.38}\left(^{+0.41}_{-0.40}\right)$ | $^{+0.07}_{-0.07}\left(^{+0.09}_{-0.05}\right)$ | $1.00^{+0.37}_{-0.35}\left(^{+0.37}_{-0.36}\right)$ |
| tH | $0.20^{+0.01}_{-0.03}$ | $1.27^{+0.76}_{-0.69}\left(^{+0.76}_{-0.20}\right)$ | $^{+0.75}_{-0.68}\left(^{+0.76}_{-0.20}\right)$ | $^{+0.10}_{-0.13}\left(^{+0.08}_{-0.08}\right)$ | $6.24^{+3.72}_{-3.37}\left(^{+3.73}_{-1.00}\right)$ |

(b) Experimental results from arXiv:2103.06956.

Figure 7: Partial $\Delta\chi^2$ and experimental results for every bins from arXiv:2103.06956.

where the red circle means $\sigma_B/(\sigma_B)_{SM} > 1$ and the blue cross means $\sigma_B/(\sigma_B)_{SM} < 1$

As we can see in Figure 7(a), when $m_{H^\pm}$ increases, the $\Gamma(\gamma\gamma)$ goes down. And luckily, we find the $\Delta\chi^2$ goes up with respect to almost every bins that $\sigma_B/(\sigma_B)_{SM} > 1$. This supports that smaller $\chi^2$ doesn't mean SM alignment.

Also, We can notice two special bins: ttH & ggH_BSM

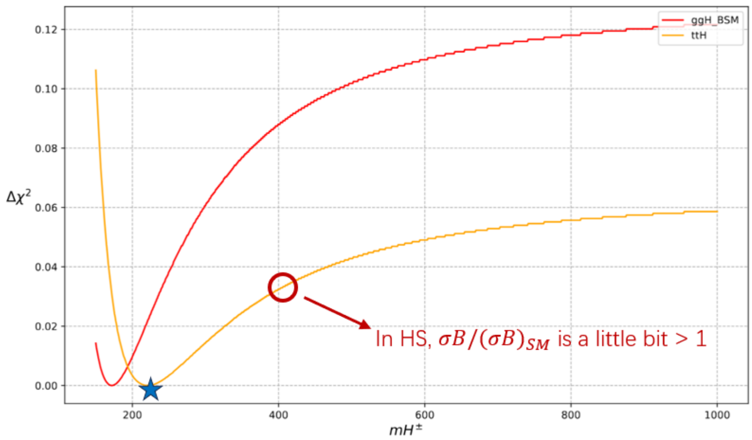



Figure 8: Two special bins: ttH and ggH_BSM.

According to Figure 8, we can see that the $\sigma_B/(\sigma_B)_{SM}$ of ttH is almost 1. Does this bin can tell which point is the global SM alignment point? Let's suppose the star point is the global SM alignment point. So, at that point the $\sigma_B/(\sigma_B)_{SM}$ of ggH_BSM is around 1. Because as $m_{H^\pm}$ decreases the $\Gamma(\gamma\gamma)$ goes up, the $\sigma_B/(\sigma_B)_{SM}$ of ggH_BSM also increases and becomes bigger than 1 and further away from O.94. But as we can see in the figure, the Ax$^2$actually becomes smaller.

All this evidents supports that the $\chi^2$ minimum is not the SM alignment limit. However, for studying alignment, we need a decoupling point.

So let's talk about theoretical decoupling from the $\Gamma(\gamma\gamma)$ formula(15):

When $m_{H^\pm}^2 \gg m_h^2$, we have $\tau_{H^\pm} \ll 1$. The loop function behaves as:

$$f(\tau_{H^\pm}) \to \tau_{H^\pm} + \frac{1}{3}\tau_{H^\pm}^2, \quad A_0(\tau_{H^\pm}) \to \frac{1}{3} \tag{22}$$

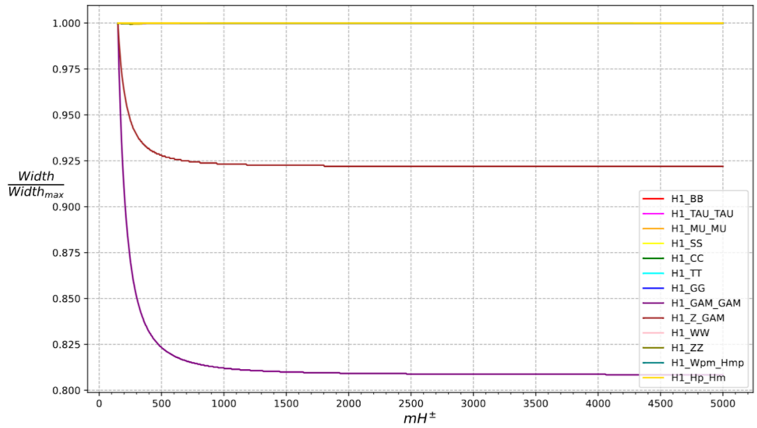



Figure 9: Two special bins: ttH and ggH_BSM.

Although $g_{hH^\pm H^\mp}$ depends on $m_{H^\pm}$, the influence can be counteracted. So as $m_{H^\pm}$ varies, it can be seen as a constant. Then effect scales as $1/m_{H^\pm}^2$, ensuring decoupling(Figure 9). We thus choose $m_{H^\pm} = 800$ GeV.

# 7 Deviations from the SM Alignment Point

## 7.1 Deviation in $\alpha_1$

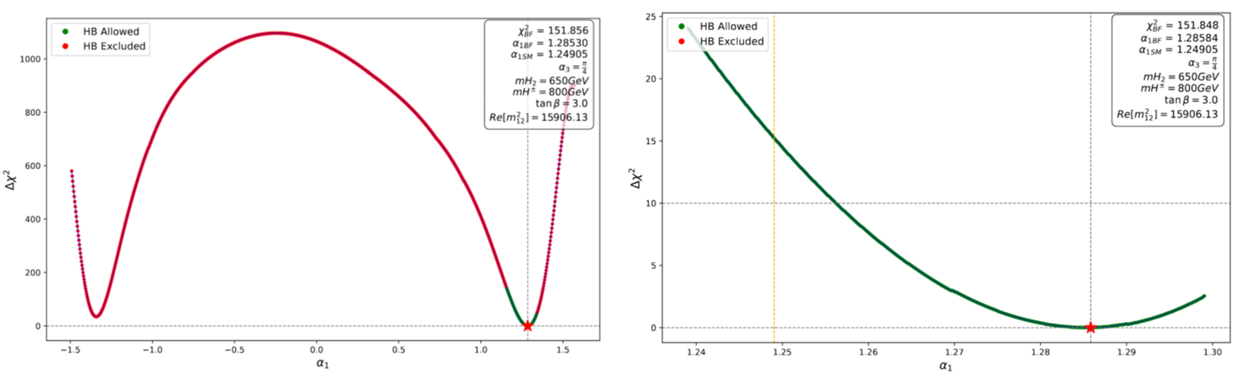
We define $\alpha_1 = \beta + \delta$. $\delta$ represents the deviation from SM alignment in $\alpha_1$.

The couplings of the SM-like Higgs $H_1$ are modified as follows:

$$c(H_1VV) = \cos\delta \tag{23}$$

$$|c(H_1b\bar{b})|^2 = \frac{\cos^2(\beta+\delta)}{\cos^2\beta} \tag{24}$$

$$|c(H_1t\bar{t})|^2 = \frac{\sin^2(\beta+\delta)}{\sin^2\beta} \tag{25}$$




(a) Global fit for $\alpha_1$ for a specific benchmark point.

(b) Detailed $\Delta\chi^2$.

Figure 10: $\alpha_1$ deviation for a specific benchmark point.

According to Figure 10, $\alpha_{1,BFP} > \alpha_{1,SM}$. For the specific benchmark point found in the scan ($\delta \approx 0.03597$):

$$\Delta|c(H_1VV)|^2 \approx -1.29 \times 10^{-3}, \quad \Delta|c(H_1b\bar{b})|^2 \approx -0.205, \quad \Delta|c(H_1t\bar{t})|^2 \approx 0.0228 \tag{26}$$

This implies: $|c(H_1VV)|^2 < 1$, $|c(H_1b\bar{b})|^2 < 1$, and $|c(H_1t\bar{t})|^2 > 1$.
And we can see,

- $\Delta\chi^2$ between $\alpha_{1,BFP}$ and $\alpha_{1,SM}$ is approximately 15.
- As for the sensitivity, when $\Delta\chi^2$ reach 10(99 % C.L.), $\delta$ is around 0.05.

Now let's show the BFP and sensitivity for different $\tan\beta$.

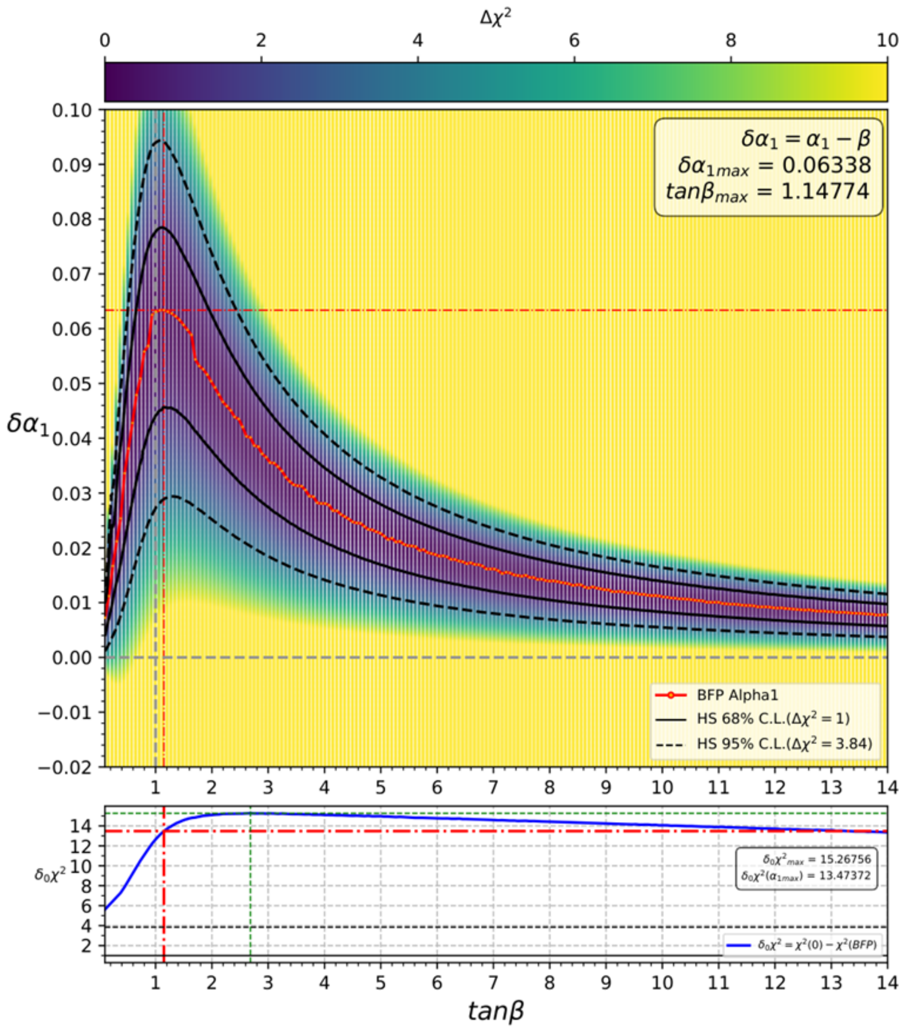



Figure 11: BFP and $\delta_0\chi^2(\chi^2_{SM} - \chi^2_{BFP})$ of $\alpha_1$ for different $\tan\beta$.

In Figure 11 we can see that as $tan\beta$ varies from around 0 to 14, $\delta\alpha_1$ is always bigger than 0. And $\delta_0\chi^2(\chi^2_{SM} - \chi^2_{BFP})$ is always $> 4$(95% C.L.)
The sensitivity of the fit strictly depends on $\tan\beta$:

- When $\tan\beta$ is around 1, the sensitivity is the lowest. This is because when $\alpha_1$ deviates from $\pi/4$, the terms $\sin\beta$ and $\cos\beta$ in the Yukawa coupling denominators play a crucial role. A deviation makes $|c(H_1b\bar{b})|^2$ or $|c(H_1t\bar{t})|^2$ change more drastically as we move away from $\tan\beta \sim 1$, thereby improving the sensitivity in those regions.

- As $\tan\beta$ becomes larger, the sensitivity increases.
- Even in the region where $\tan\beta$ is close to 1 (lowest sensitivity), a small change of about $\delta \approx 0.03$ in $\alpha_1$ still results in a $\Delta\chi^2$ of about 4 (95% C.L.).

### 7.2 Deviation in $\alpha_2$

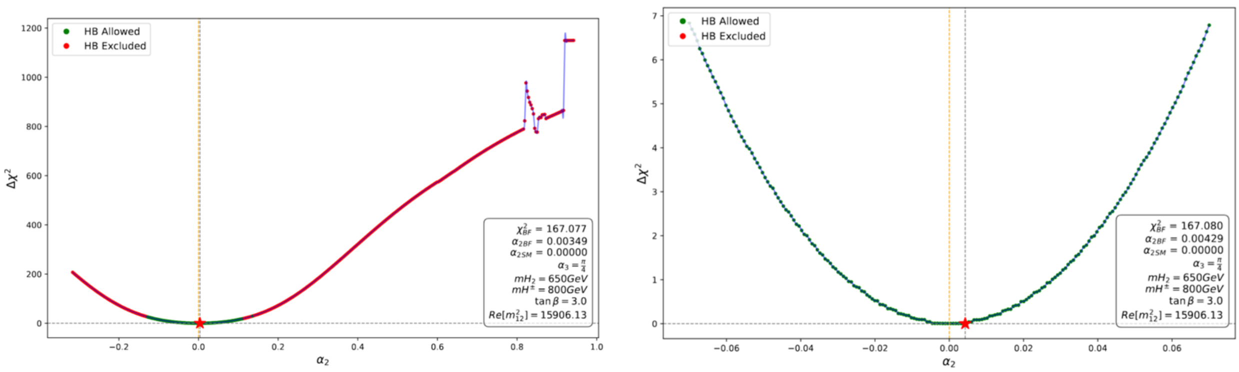
We define $\alpha_2 = \delta$. $\delta$ represents the deviation from SM alignment in $\alpha_2$.

The couplings of the SM-like Higgs $H_1$ are modified as follows:

$$c(H_1VV) = \cos\delta \tag{27}$$

$$|c(H_1b\bar{b})|^2 = \cos^2\delta + \tan^2\beta\sin^2\delta \tag{28}$$

$$|c(H_1t\bar{t})|^2 = \cos^2\delta + \frac{\sin^2\delta}{\tan^2\beta} \tag{29}$$




(a) Global fit for $\alpha_2$ for a specific benchmark point.

(b) Detailed $\Delta\chi^2$.

Figure 12: $\alpha_2$ deviation for a specific benchmark point.

According to Figure 12, $\alpha_{2,BFP} > \alpha_{2,SM}$. For the specific benchmark point found in the scan ($\delta \approx 0.00429$):

$$\Delta|c(H_1VV)|^2 \approx -1.84\times10^{-5}, \quad \Delta|c(H_1b\bar{b})|^2 \approx 1.47\times10^{-4}, \quad \Delta|c(H_1t\bar{t})|^2 \approx -1.64\times10^{-5} \tag{30}$$

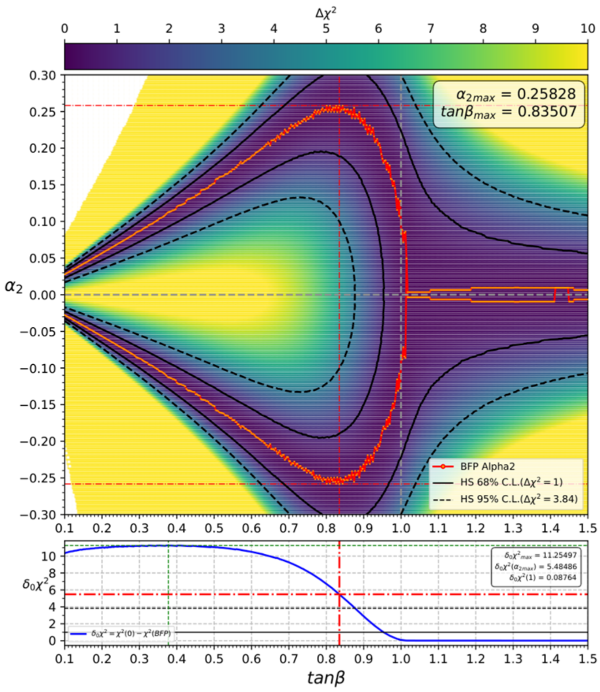
This implies: $|c(H_1VV)|^2 < 1$, $|c(H_1b\bar{b})|^2 > 1$, and $|c(H_1t\bar{t})|^2 < 1$ (for $\tan\beta > 1$).

And we can see,

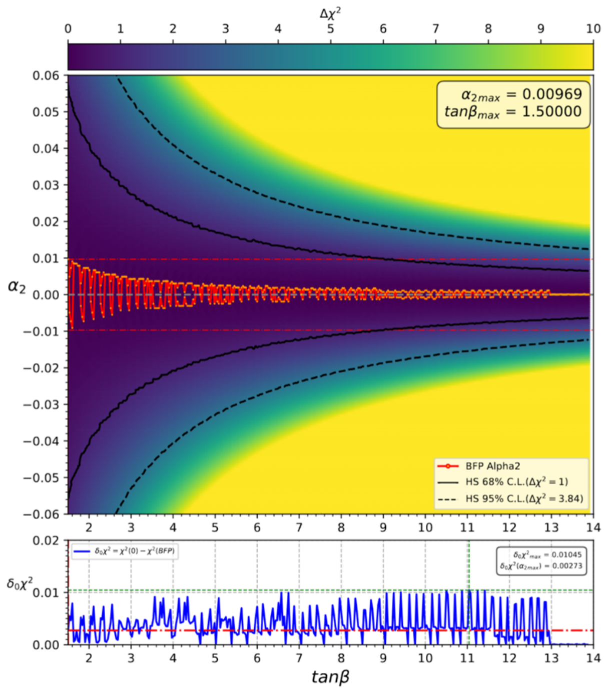
- $\Delta\chi^2$ between $\alpha_{2,BFP}$ and $\alpha_{2,SM}$ is nearly 0.
- As for the sensitivity, when $\Delta\chi^2$ reach 4(95 % C.L.), $\delta$ is around 0.05.

Then we analyze the global fit results for $\alpha_2$ across different $\tan\beta$ regions, as illustrated in the scan plots (see Figure 13):

- Region $\tan\beta < 1$: As $\tan\beta$ decreases below 1, the Best Fit Point ($\alpha_{2,BFP}$) significantly deviates from 0 (the CP-conserving limit). The improvement in the fit, represented by $\delta_0\chi^2$ (defined as $\chi^2(SM) - \chi^2_{BFP}$), can reach approximately 10 (99% C.L.). This suggests that in the low $\tan\beta$ regime, the data may favor a CP-mixed scalar state.
- Region $\tan\beta > 1$: As $\tan\beta$ increases above 1, $\alpha_{2,BFP}$ rapidly converges back to $\sim 0$, and $\delta_0\chi^2$ becomes nearly 0, consistent with the SM alignment.




(a) $0.1 < \tan\beta < 1.5$.




(b) $1.5 < \tan\beta < 14$.

Figure 13: BFP and $\delta_0\chi^2(\chi^2_{SM} - \chi^2_{BFP})$ of $\alpha_2$ for different $\tan\beta$.

The sensitivity of the $\chi^2$ fit to deviations in $\alpha_2$ shows a strong dependence on $\tan\beta$ too:

- Minimum Sensitivity ($\tan\beta \sim 1$): Similar to the $\alpha_1$ case, the sensitivity decreases significantly when $\tan\beta$ is around 1. That's because when $tan\beta = 1$, $|c(H_1 b\bar{b})|^2$ or $|c(H_1 t\bar{t})|^2$ will be 1 constantly, thereby reducing the sensitivity in those regions.
- Low $\tan\beta$ Region ($\tan\beta < 0.8$): The sensitivity increases apparently as $\tan\beta$ decreases. In this region, a deviation of about $\delta \approx 0.05$ in $\alpha_2$ can result in a $\Delta\chi^2$ of approximately 4 ($\sim 95\%$ C.L.). This enhanced sensitivity at large $\tan\beta$ is driven by the $|c(H_1 t\bar{t})|^2$ term, which contains a $1/t_\beta^2$ enhancement factor.
- High $\tan\beta$ Region: The sensitivity goes up significantly as $\tan\beta$ increases. In the large $\tan\beta$ region, the constraints are tighter; a smaller change of about $\delta \approx 0.02$ in $\alpha_2$ is sufficient to result in a $\Delta\chi^2$ of approximately 4 ($\sim 95\%$ C.L.). This enhanced sensitivity at large $\tan\beta$ is driven by the $|c(H_1 b\bar{b})|^2$ term, which contains a $t_\beta^2$ enhancement factor.

# 8 Conclusion and Further Research

## 8.1 Conclusion

Currently, we mainly consider the constraints from current 125 GeV Higgs data. The off-shell decay and heavy Higgs data are not included yet, which may further constrain the mass of other scalar particles and can be done by HiggsBounds.

- For current 125 GeV Higgs data, it's insensitive to $\alpha_3$(CP property of $H_{2\&3}$) and $m_{H_{2\&3}}$, but $m_{H^\pm}$ significantly influences the $H_1 \to \gamma\gamma/Z\gamma$ decay width via loop effects.
- The BFP for $\alpha_1$ does not strictly coincide with $\beta(\alpha_{1BF} > \alpha_{1SM})$, suggesting the experimental data has a slight deviation from the SM alignment limit.

- As for $\alpha_2$: $\tan\beta > 1$, the BFP is located near 0, $H_1$(125 GeV SM-like) $\to$ a CP-even scalar. But $\tan\beta < 1$, $\alpha_2$ has a significant deviation from SM alignment point, $H_1$(125 GeV SM-like) $\to$ a CP-mixing scalar

- Thirdly, the sensitivity of $\Delta\chi^2$ to $\alpha_1$ and $\alpha_2$ highly depends on $\tan\beta$. Generally, the sensitivity becomes lower when $\tan\beta$ is around 1, and it increase as $\tan\beta$ goes up.

### 8.2 Further Research

- Firstly, currently we can only provide HiggsSignals with inclusive cross sections and branch ratios, and it uses the SM Higgs boson as a kinematic template to divide inclusive cross sections and branch ratios into different bins. So theoretically we should use MC simulation to get the kinematic distribution in C2HDM to get the modification factors for STXS bins.

- Secondly, we should add more theoretical constrains and experimental constrains to get a benchmark parameters region. And use MC stimulation to do further research to find the CP-violation signals.

## Acknowledgments

I would like to express my sincere gratitude to my supervisor, Prof. Yusheng Wu, for his meticulous guidance, patient mentorship, and valuable insights throughout this research. His support has been instrumental in shaping this project.

I am also deeply grateful to the judges of the Undergraduate Research Program defense for their time, hard work, and constructive feedback, which have greatly contributed to my academic growth.

Finally, I would like to thank the University of Science and Technology of China (USTC) for providing the Undergraduate Research Program, an excellent platform that has offered me this invaluable opportunity to engage in scientific research.

## References

[1] P. Posch, *Enhancement of $h \to \gamma\gamma$ in the Two Higgs Doublet Model Type I*, Phys. Lett. B **696**, 447–453 (2011).

[2] D. Fontes, M. Muhlleitner, J. C. Romao, et al., *The C2HDM revisited*, JHEP **02**, 073 (2018) [arXiv:1711.09419].

[3] H. Bahl, et al., *HiggsTools: BSM scalar phenomenology with new versions of HiggsBounds and HiggsSignals*, Comput. Phys. Commun. **291**, 108803 (2023) [arXiv:2210.09332].

[4] ATLAS Collaboration, *Measurements of Higgs boson production cross-sections in the $H \to \gamma\gamma$ decay channel in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector*, JHEP **08**, 005 (2021) [arXiv:2103.06956].